Practice Exercises: Teaching Block 4

- Numbers
- Exception Handling
- File I/O
- Fill in the Gaps

This set of exercises is in addition to those included directly in lecture slides (and extra reading materials), which you should also attempt.

Question 1

- Consider the following Java program:

```
import java.util.*;
public class RandomNums {
  public static void main(String args[]) {
    System.out.print("How many random nums?");
    Scanner inputScanner = new Scanner(System.in);
    int n = inputScanner.nextInt();
    for (int i=0; i<n; i++) {
      System.out.println("Random num " + i + " is " +
                                Math.random());
    }
    inputScanner.close();
  }
}
```

1. Explain what this program does.
2. What will be the output if the user enters 3?
3. Briefly describe what `Math.random()` returns.

Question 2

- Identify the constructors in class **SonOfBoo** that are not legal. **Explain** your answer.

```
public class Boo {
  public Boo(int i) { }                    (1)
  public Boo(String s) { }                 (2)
  public Boo(String s, int i) { }          (3)
}

class SonOfBoo extends Boo {
  public SonOfBoo() { super("boo"); }
  public SonOfBoo(int i) { super("Fred"); }
  public SonOfBoo(String s) { super(42); }
  public SonOfBoo(String a, String b, String c) { super(a, b); }
  public SonOfBoo(int i, int j) { super("man", j); }
  public SonOfBoo(int i, int x, int y) { super(i, "star"); }
}
```

Question 3

- What is wrong with the code? **Justify** your answer.

ExampleGoneWrong

```
public class IS
{
    public int i = 0;
    public static int s = 0;
    public void doSomethingInstance() {
        System.out.println("Instance Method");
        System.out.println(i);
        System.out.println(s);
    }
    public static void doSomethingStatic() {
        System.out.println("Static Method");
        System.out.println(i);
        System.out.println(s);
    }
    public static void main(String args[]){
        doSomethingStatic();
        doSomethingInstance();
    }
}
```

Question 4

- Consider a command-line calculator application. Write a program to deal with non-numeric operands, so it displays a message telling the user of the wrong operand type before exiting using an *exception handler*.

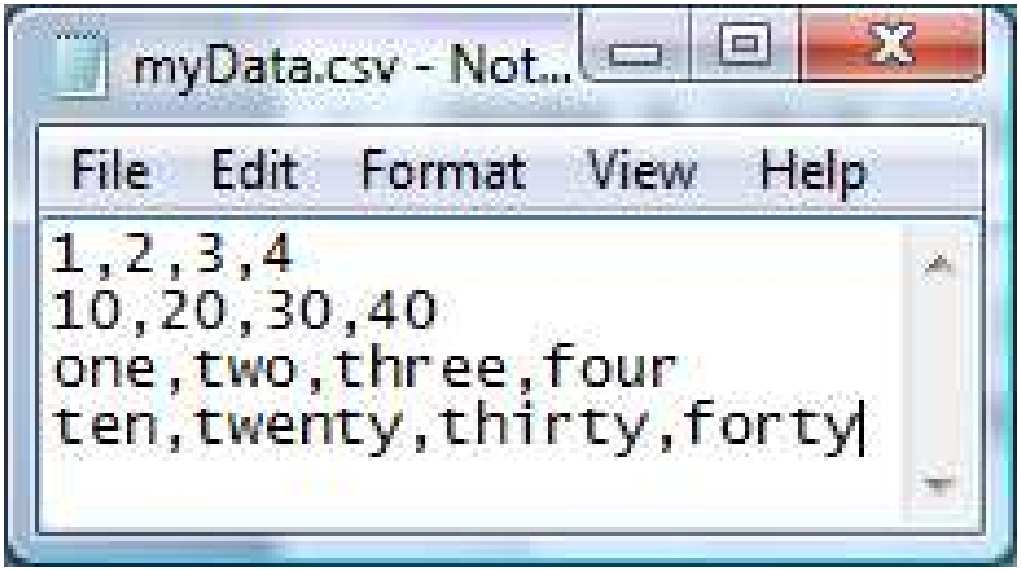
```
public class Calculator {
  public static void main(String[] args) {
    if (args.length != 3) {
      System.out.println("Usage: java Calculator op1 operator op2");
      System.exit(0);
    }
    int result = 0;
    switch (args[1].charAt(0)) {
      case '+': result = Integer.parseInt(args[0]) +
                         Integer.parseInt(args[2]); break;
      case '-': result = Integer.parseInt(args[0]) -
                         Integer.parseInt(args[2]); break;
      // Similarly for the '*' and '/' operations ...
    }
    System.out.println(args[0] + args[1] + args[2] + " = " + result);
  }
}
```

Can you think of other potential problems with this program?

Question 5

- Write a Java program that reads each line of the data in a file called **`myData.csv`** and stores it in an array, after which it *displays the first element of the array onto the console* for each line of the file. Below is the expected output for running the program with the file shown:

```
> java FileProcessor
1st element of line 1 = 1
1st element of line 2 = 10
1st element of line 3 = one
1st element of line 4 = ten
```



Question 6

- What is the output? Explain your answer.

```java
public class FinalPass {
    final static int[] array1 = new int[] {0, 1};
    final static int[] array2 = new int[] {1, 0};

    public static void main(String[] args) {
        System.out.println(array1);
        System.out.println("A: " + array1[0] + ", " + array1[1]);

        array1=array2;
        System.out.println(array1);
        System.out.println("B: " + array1[0] + ", " + array1[1]);

        array1[0]++;
        System.out.println(array1);
        System.out.println("C: " + array1[0] + ", " + array1[1]);

        array1[0] = array2[0];
        array1[1] = array2[1];
        System.out.println(array1);
        System.out.println("D: " + array1[0] + ", " + array1[1]);

        method1(array1);
        System.out.println(array1);
        System.out.println("F: " + array1[0] + ", " + array1[1]);
    }

    public static void method1(int[] array){
        System.out.println(array);
        System.out.println("E: " + array1[0] + ", " + array1[1]);
        array[0] = 9;
        array[1] = 9;
    }
```

Exercise: Fill in the gaps

```
public class InvalidAgeException ____(1)_____ {
  public InvalidAgeException() {
    super("Invalid Age Was Entered...");
  }
}
```

(1) The InvalidAgeException class needs to extend the _____ class.
(2) The PersonAge class uses the Scanner class; therefore, we are required to _____.

```
import java.io.*;  ____(2)_____;
public class PersonAge {
  int age;
  AgeValidity myAge;
  Scanner input;
  public void checkAge() throws InvalidAgeException {
    input = new Scanner(System.in);
    myAge = new AgeValidity();
    System.out.println("Please enter your age : ");
    age = input.nextInt();
    if (!myAge.checkAgeValidity(age))
      ____(3)_____ new InvalidAgeException();
  }
}
```

(3) The _____ keyword needs to be used before the InvalidAgeException class name to show the occurrence of the error.